

Wednesday 09/01/2020

Final Exam

Duration: 90 minutes

Name:

Student No:

P1 [40 points]

1. Indicate whether the following are True or False by circling T or F.

$\overline{A_{TM}}$ is mapping reducible to A_{TM}	T	F
$\overline{A_{TM}}$ is Turing reducible to A_{TM}	T	F
If A is decidable relative to B , then A is Turing reducible to B .	T	F
If $A \leq_T B$ and $B \leq_T C$, then $A \leq_T C$	T	F
If $A \leq_T B$, and A is decidable, then B is decidable.	T	F
Given $A = 01010101010101010101010101010101$ and $B = 1110010110100011101010000111010011010111$, B contains more information compared to A .	T	F
We can say that if a string is more compressible, it is more random	T	F
Solutions of NP problems always require exponential time.	T	F
$\log^2 n = O(n)$	T	F
$n^2 = O(n \log n)$	T	F

2. Fill in the blanks with these: P, NP, CFL, RL, T.Decidable, T.Recognizable:

$\subseteq$ $\subseteq$ $\subseteq$ $\subseteq$ $\subseteq$

P2 [20 points] Recall that a **vertex cover** is a subset of vertices that covers (touches) all the edges in a graph. Formally, given an undirected graph $G = (V, E)$, a vertex cover is a subset $V' \subseteq V$ such that if $(i, j) \in E$, then either $i \in V'$ or $j \in V'$ (or both). Then the decision problem, VERTEX-COVER, is to determine whether a graph has a vertex cover of a given size k . It is written in language form as follows: $VERTEX-COVER = \{\langle G, k \rangle \mid G \text{ is an undirected graph that has a } k\text{-node vertex cover}\}$.

A **clique**, on the other hand, is a subset of vertices that are all directly connected. Formally, given an undirected graph $G = (V, E)$, a clique is a subset $V' \subseteq V$ of vertices, each pair of which is connected by an edge in E . $CLIQUE = \{\langle G, k \rangle \mid G \text{ is an undirected graph that has a } k\text{-node clique}\}$.

Prove that CLIQUE is NP-Complete, by using the fact that VERTEX-COVER is NP-complete.

P3 [20 points] DENSE-SUBGRAPH: Given a graph G and two integers m and n , does G have a set of m vertices with at least n edges between them? Prove that DENSE-SUBGRAPH is NP-complete.

P4 [20 points] SUBSET-SUM: Given a set S of n non-negative integers, does there exist a subset of k of the integers that sums to a value c ? We know that SUBSET-SUM is NP-complete.

HALF-PARTITION: Given a set S of n non-negative integers, does there exist a partition of S into subsets X and Y such that $X \cup Y = S$, $X \cap Y = \emptyset$ the sum of the integers in X equals the sum of the integers in Y ?

Prove that HALF-PARTITION is NP-complete.

P5 (Bonus) [20 points] A **coloring** of a graph is an assignment of colors to its nodes so that no two adjacent nodes are assigned the same color. Let $3COLOR = \{\langle G \rangle \text{ is colorable with 3 colors}\}$. Show that $3COLOR$ is NP-complete. (Hint: Use the following three subgraphs.)

